

Calculus: Concept Checklist

Active-Recall & Self-Testing Revision Guide

1 Functions, Models, and Transformations

▮ Conventions & Exam Pitfalls.

- **Horizontal vs. Vertical Scaling:** For $c > 1$, $f(cx)$ is a horizontal ***compression*** by factor c , whereas $cf(x)$ is a vertical ***stretch*** by factor c .
- **Trigonometric Periods:** The fundamental period of $\sin x$ is 2π , while the period of $\sin(ax)$ ($a > 0$) is compressed to $\frac{2\pi}{a}$.

▮ Definitions & Concepts to State 1.1

- **Domain and Range:** Define the domain D and image (range) $\text{Im}(f) = \{f(x) : x \in D\}$ of a real-valued function f .
- **Monotonicity:** Define strictly increasing, strictly decreasing, and strictly monotonic functions.
- **Function Classifications:** Define the following standard families of functions:
 - Power functions x^α (polynomial for $\alpha = n \in \mathbb{Z}^+$, root for $\alpha = 1/n$, reciprocal for $\alpha = -1$).
 - Rational functions $R(x) = P(x)/Q(x)$ where P, Q are polynomials.
 - Algebraic functions (functions constructed using finite algebraic operations and roots).
 - Periodic functions ($f(x+p) = f(x)$ for all x , fundamental period $p > 0$).
 - Exponential functions $f(x) = a^x$ ($a > 0, a \neq 1$).
- **Horizontal Line Test:** State the Horizontal Line Test for function injectivity (one-to-one property).
- **Graph Transformations:** For $c > 0$, state the geometric effect on $y = f(x)$ of:

$$y = cf(x), \quad y = \frac{1}{c}f(x), \quad y = f(cx), \quad y = f\left(\frac{x}{c}\right).$$

¶ Theorems & Identities to State 1.2

- **Composite Periodicity:** If f has fundamental period p , then $g(x) = f(ax)$ has fundamental period p/a for any $a > 0$.

¶ Proof Challenges & Calculations 1.3

- **Period Derivation:** Prove that the fundamental period of $f(x) = \sin(ax)$ is $\frac{2\pi}{a}$ using the definition $f(x+p) = f(x)$.

2 Limits, Epsilon-Delta Proofs, and Continuity

¶ Conventions & Exam Pitfalls.

- **Horizontal Asymptotes Requirement:** When finding horizontal asymptotes, you must evaluate *both* directional limits: $x \rightarrow +\infty$ and $x \rightarrow -\infty$.
- **ϵ - δ Bounding Strategy:** In quadratic limit proofs $\lim_{x \rightarrow a} f(x) = L$, choose an initial bound (e.g., $\delta_1 = 1$) to bound the extra factor $|x + c| \leq M$, then set $\delta = \min \left\{ 1, \frac{\epsilon}{M} \right\}$.

Definitions & Concepts to State 2.1

- **Precise Limit Definition (ϵ - δ):** State the formal (ϵ, δ) -definition of $\lim_{x \rightarrow a} f(x) = L$:

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \implies |f(x) - L| < \epsilon.$$

- **One-Sided Limits:** State the (ϵ, δ) -definitions of the left-hand limit $\lim_{x \rightarrow a^-} f(x) = L$ and right-hand limit $\lim_{x \rightarrow a^+} f(x) = L$.
- **Infinite Limits & Asymptotes:**
 - Define infinite limit $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} f(x) = -\infty$ (vertical asymptote $x = a$).
 - Define limits at infinity $\lim_{x \rightarrow \infty} f(x) = L$ and $\lim_{x \rightarrow -\infty} f(x) = L$ (horizontal asymptote $y = L$).
- **Continuity Definitions:** State what it means for f to be:
 - Continuous at a point $x = a$ ($\lim_{x \rightarrow a} f(x) = f(a)$).
 - Continuous from the left / right at $x = a$.
 - Continuous on an open interval (a, b) and on a closed interval $[a, b]$.

Theorems & Identities to State 2.2

- **Composite Limit Theorem:** If $\lim_{x \rightarrow a} g(x) = L$ and f is continuous at L , then $\lim_{x \rightarrow a} f(g(x)) = f(L)$.
- **Differentiability Implies Continuity:** If f is differentiable at $x = a$, then f is continuous at $x = a$. (State why the converse is false with $f(x) = |x|$).
- **Sign-Preserving / Separation Lemma:** If $\lim_{x \rightarrow a} g(x) = B \neq 0$, then there exists a positive number $\delta > 0$ such that:

$$0 < |x - a| < \delta \implies |g(x)| > \frac{|B|}{2}.$$

□ Proof Challenges & Calculations 2.3

□ Interval Continuity Proofs:

1. Show that $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on $[-1, 1]$ (including one-sided endpoints).
2. Show that $h(x) = |x^2 - 3x - 6|$ is continuous at every real number.

□ **Differentiability $\implies$ Continuity Proof:** Prove that if $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists, then $\lim_{x \rightarrow a} f(x) = f(a)$.

□ **Separation Lemma Proof:** Prove the Separation Lemma using the formal ϵ - δ limit definition with $\varepsilon = \frac{|B|}{2}$.

□ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 2.4	Problems 39, 43	□ Done

3 Differentiation, Special Trigonometric Limits, and Differentials

□ Definitions & Concepts to State 3.1

- **Derivative Definition:** State the definition of the derivative $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.
- **Chain Rule Formula:** State the Chain Rule for composite functions: $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$.
- **Differentials vs. Increments:** Distinguish among:
 - Independent variable increment Δx and differential dx ($dx = \Delta x$).
 - Dependent variable true increment $\Delta y = f(x + \Delta x) - f(x)$.
 - Dependent variable differential $dy = f'(x)dx$ and the linear approximation $\Delta y \approx dy$.

▮ Theorems & Identities to State 3.2

▫ **Fundamental Trigonometric Limits:**

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

- **Chain Rule Theorem:** If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ is differentiable at x with $F'(x) = f'(g(x))g'(x)$.

▮ Proof Challenges & Calculations 3.3

- **Trigonometric Limits Evaluation:** Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$ algebraically using half-angle identities and the squeeze theorem.
- **Chain Rule Proof:** Write the full formal proof of the Chain Rule using the differentiability error formulation $\Delta y = f'(a)\Delta u + \varepsilon\Delta u$.
- **Differential Error Estimation:** Calculate dy versus Δy for $y = x^3 + 2x$ as x changes from 2 to 2.05.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 3.5	Problem 75	▫ Done
Section 3.9	Problems 35, 42	▫ Done
Section 3.10	Problems 33, 35, 40	▫ Done

4 Applications of Derivatives, Mean Value Theorem, and Curve Sketching

▮ Definitions & Concepts to State 4.1

▫ **Extrema and Critical Points:**

- Define absolute (global) maximum/minimum and local (relative) maximum/minimum.
- Define a **critical point** ($f'(c) = 0$) and a **singular point** ($f'(c)$ undefined).

▫ **Rolle's Theorem & Mean Value Theorem:** State the hypotheses and conclusions of:

- Rolle's Theorem
- Mean Value Theorem (MVT)

▫ **Concavity and Inflection Points:** Define concave up, concave down, and inflection point in terms of $f'(x)$ and $f''(x)$.

▫ **8-Step Curve Sketching Guide:** List all 8 standard systematic steps for sketching the graph of $y = f(x)$:

1. Domain
2. Intercepts (x and y intercepts)
3. Symmetry (Even, Odd, Periodic)
4. Asymptotes (Horizontal, Vertical, Slant)
5. Intervals of Increase and Decrease (First Derivative Test)
6. Local Extrema
7. Concavity and Inflection Points (Second Derivative Test)
8. Sketch the Curve

▫ **Newton-Raphson Method:** State the iteration formula for finding roots:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

▮ Theorems & Identities to State 4.2

- **Extreme Value Theorem (EVT):** If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum and an absolute minimum on $[a, b]$.
- **Fermat's Theorem (Critical Point Theorem):** If f has a local extremum at c and $f'(c)$ exists, then $f'(c) = 0$.
- **Constant Function Theorem:** If $F'(x) = 0$ for all $x \in (a, b)$, then $F(x) = C$ is constant on (a, b) .
- **Equal Derivatives Theorem:** If $f'(x) = g'(x)$ for all $x \in (a, b)$, then $f(x) = g(x) + C$ on (a, b) .

▮ Proof Challenges & Calculations 4.3

- **Uniqueness of Real Root:** Prove that the cubic equation $x^3 + x - 1 = 0$ has *strictly one real root* using the Intermediate Value Theorem (existence) and Rolle's Theorem / MVT (uniqueness).
- **Zero Derivative Characterization Proof:** Prove that if $F'(x) = 0$ on (a, b) , then $F(x) = C$ using the Mean Value Theorem.
- **Newton-Raphson Derivation:** Derive the Newton-Raphson iteration formula geometrically from the tangent line approximation at $(x_n, f(x_n))$.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 4.2	Problems 19, 25, 29	▫ Done
Section 4.3	Problems 75, 83	▫ Done
Section 4.4	Problem 55	▫ Done
Section 4.5	Problem 73	▫ Done
Section 4.8	Problem 17	▫ Done

5 Definite Integrals and Fundamental Theorems of Calculus

▮ Definitions & Concepts to State 5.1

- **Riemann Sums Definition:** Define the Riemann sum $\sum_{i=1}^n f(x_i^*)\Delta x$ of f on $[a, b]$ where $\Delta x = \frac{b-a}{n}$. Define Left sum ($x_i^* = x_{i-1}$), Right sum ($x_i^* = x_i$), and Midpoint sum ($x_i^* = \bar{x}_i$).
- **Definite Integral Definition:** Define $\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)\Delta x$.
- **Net Displacement vs. Distance Traveled:**
 - Net displacement = $\int_{t_1}^{t_2} v(t)dt$.
 - Total distance traveled = $\int_{t_1}^{t_2} |v(t)|dt$.

▮ Theorems & Identities to State 5.2

- **Integrability of Continuous Functions:** If f is continuous on $[a, b]$, or bounded with only finitely many jump discontinuities, then f is integrable on $[a, b]$.
- **Comparison and Boundedness Properties:**
 - If $f(x) \leq g(x)$ on $[a, b]$, then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$.
 - If $m \leq f(x) \leq M$ on $[a, b]$, then $m(b-a) \leq \int_a^b f(x)dx \leq M(b-a)$.
- **First Fundamental Theorem of Calculus (FTC 1):** Let f be continuous on $[a, b]$. If $g(x) = \int_a^x f(t)dt$ for $x \in [a, b]$, then g is continuous on $[a, b]$, differentiable on (a, b) , and:

$$g'(x) = \frac{d}{dx} \int_a^x f(t)dt = f(x).$$

- **Second Fundamental Theorem of Calculus (FTC 2):** If f is continuous on $[a, b]$ and F is any antiderivative of f ($F' = f$), then:

$$\int_a^b f(x)dx = F(b) - F(a) = [F(x)]_a^b.$$

- **Symmetric Integral Theorem:** For an integrable function f on $[-a, a]$:

$$\int_{-a}^a f(x)dx = \begin{cases} 2 \int_0^a f(x)dx & \text{if } f \text{ is even } (f(-x) = f(x)) \\ 0 & \text{if } f \text{ is odd } (f(-x) = -f(x)) \end{cases}$$

▮ Proof Challenges & Calculations 5.3

- **FTC 1 Proof:** Write the complete proof of FTC 1 using the definition of derivative and the Extreme Value Theorem / Boundedness property.
- **FTC 2 Proof:** Prove FTC 2 using FTC 1 and the Constant Function Theorem.
- **Integral Estimation Task:** Give rigorous upper and lower bounds for $\int_0^1 e^{-x^2} dx$ using the boundedness property on $[0, 1]$.

▮ **Textbook Exercise Checklist.**

Section	Textbook Problems	Status
Section 5.1	Problem 5	□ Done
Section 5.2	Problems 55, 57	□ Done
Section 5.3	Problems 13, 17, 75	□ Done
Section 5.4	Problems 45, 59, 61	□ Done
Section 5.5	Problems 61, 67, 73, 91, 93	□ Done

6 Applications of Definite Integrals: Volumes and Average Values

▮ **Definitions & Concepts to State 6.1**

□ **Solids of Revolution Methods:**

- **Disk Method:** $V = \pi \int_a^b [R(x)]^2 dx$ (cross-sections perpendicular to axis of revolution).
- **Washer Method:** $V = \pi \int_a^b ([R(x)]^2 - [r(x)]^2) dx$ (outer radius $R(x)$, inner radius $r(x)$).
- **Cylindrical Shells Method:** $V = 2\pi \int_a^b (\text{shell radius})(\text{shell height}) dx$ (cylindrical shells parallel to axis of revolution).

- **Average Value of a Function:** State the definition of the average value f_{avg} of f on $[a, b]$:

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

▮ **Theorems & Identities to State 6.2**

- **Mean Value Theorem for Integrals:** If f is continuous on $[a, b]$, then there exists a number $c \in [a, b]$ such that:

$$f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx \Leftrightarrow \int_a^b f(x) dx = f(c)(b-a).$$

▮Proof Challenges & Calculations 6.3

- **Washer vs. Shell Setup:** Set up definite integrals to find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = 2x$ about:
 1. The x -axis (Washer method).
 2. The line $x = 3$ (Cylindrical shells method).
- **MVT for Integrals Proof:** Prove the Mean Value Theorem for Integrals using the Extreme Value Theorem and the Boundedness Property of definite integrals.

▮Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 6.1	Problems 3, 15	▫ Done
Section 6.3	Problems 5, 7, 17, 19	▫ Done
Section 6.5	Problems 3, 5, 7	▫ Done

7 Techniques of Integration, Numerical Rules, and Improper Integrals

▮ Definitions & Concepts to State 7.1

- **Integration by Parts:** State the formula $\int u dv = uv - \int v du$. Outline the Tabular (DI) method.
- **Trigonometric Product-to-Sum & Sum-to-Product Formulas:**

$$\begin{aligned}\sin A \cos B &= \frac{1}{2} [\sin(A + B) + \sin(A - B)], \\ \cos A \cos B &= \frac{1}{2} [\cos(A + B) + \cos(A - B)], \\ \sin A + \sin B &= 2 \sin\left(\frac{A + B}{2}\right) \cos\left(\frac{A - B}{2}\right), \\ \cos A - \cos B &= -2 \sin\left(\frac{A + B}{2}\right) \sin\left(\frac{A - B}{2}\right).\end{aligned}$$

- **Numerical Integration Rules:**

– Trapezoidal Rule:

$$T_n = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)].$$

– Simpson's Rule (n even):

$$S_n = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)].$$

- **Improper Integrals Classifications:**

- Type 1 (Infinite intervals): $\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$.
- Type 2 (Discontinuous integrands): If f is discontinuous at b , $\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$.

▮ Theorems & Identities to State 7.2

▫ **p -Integral Convergence Test:**

$$\int_1^{\infty} \frac{1}{x^p} dx \text{ converges} \Leftrightarrow p > 1 \quad \text{and} \quad \int_0^1 \frac{1}{x^p} dx \text{ converges} \Leftrightarrow p < 1.$$

▫ **Direct Comparison Test for Improper Integrals:** Let $f(x) \geq g(x) \geq 0$ on $[a, \infty)$.

- If $\int_a^{\infty} f(x) dx$ converges, then $\int_a^{\infty} g(x) dx$ converges.
- If $\int_a^{\infty} g(x) dx$ diverges, then $\int_a^{\infty} f(x) dx$ diverges.

▮ Proof Challenges & Calculations 7.3

- **Product-to-Sum Derivations:** Derive $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$ using the angle addition formulas for sine.
- **Simpson's Rule Derivation:** Derive Simpson's rule by integrating the interpolating parabola through three equally spaced points $(-\Delta x, y_0), (0, y_1), (\Delta x, y_2)$.
- **Improper Integral Evaluation:** Determine whether $\int_0^{\infty} \frac{1}{1+x^2} dx$ and $\int_0^1 \frac{1}{\sqrt{x}} dx$ converge, and evaluate their exact values.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 7.1	Problems 9, 15, 17, 33, 41, 61, 63, 65	▫ Done
Section 7.2	Problem 41	▫ Done
Section 7.3	Problems 13, 29, 37	▫ Done
Section 7.4	Problems 15, 29, 31, 47, 51	▫ Done
Section 7.5	Problem 31	▫ Done
Section 7.7	Problem 7	▫ Done
Section 7.8	Problems 5, 13, 23, 31, 49, 51	▫ Done

8 Further Applications: Arc Length, Surface Area, and Centroids

▮ Definitions & Concepts to State 8.1

▫ Arc Length Formulas:

- Cartesian curve $y = f(x)$ on $[a, b]$: $L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$.
- Parametric curve $x = f(t), y = g(t)$ on $[\alpha, \beta]$: $L = \int_\alpha^\beta \sqrt{[f'(t)]^2 + [g'(t)]^2} dt$.
- Arc length function: $s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} dt$.

▫ Surface Area of Revolution:

- Rotation about x -axis: $S = 2\pi \int_a^b y ds = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$.
- Rotation about y -axis: $S = 2\pi \int_a^b x ds = 2\pi \int_a^b x \sqrt{1 + [f'(x)]^2} dx$.

▫ Center of Mass and Centroids:

- Discrete system of n masses: $\bar{x} = \frac{\sum m_i x_i}{\sum m_i}$.
- Planar lamina bounded by $y = f(x)$ and $y = g(x)$ ($f \geq g$):

$$A = \int_a^b [f(x) - g(x)] dx,$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx,$$

$$\bar{y} = \frac{1}{A} \int_a^b \frac{[f(x)]^2 - [g(x)]^2}{2} dx.$$

▮ Theorems & Identities to State 8.2

- **Theorem of Pappus for Volume:** Let R be a plane region rotated about an external axis in its plane. The volume of the resulting solid is $V = 2\pi \bar{r} A$, where A is the area of R and $\bar{r}$ is the distance from the axis to the centroid of R .
- **Theorem of Pappus for Surface Area:** The area of the surface generated by revolving a curve of length L about an external axis is $S = 2\pi \bar{r} L$, where $\bar{r}$ is the distance from the axis to the centroid of the curve.

▮ Proof Challenges & Calculations 8.3

- **Centroid Derivation for Planar Lamina:** Derive the integral formulas for $\bar{x}$ and $\bar{y}$ by partitioning the region into vertical approximating rectangles of mass $dm = \rho(f(x) - g(x))dx$.
- **Torus Volume and Surface Area via Pappus:** Use the Theorems of Pappus to find the volume and surface area of a torus obtained by rotating a circle of radius r centered at $(R, 0)$ ($R > r$) about the y -axis.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 8.1	Problems 9, 19	▫ Done
Section 8.2	Problems 9, 13, 17	▫ Done
Section 8.3	Problems 23, 27, 31, 45	▫ Done